AQUÍ VA EL TÍTULO DEL ARTÍCULO

JUAN CARLOS MONTER CORTÉS AND LUIS ÁNGEL ZALDÍVAR CORICHI

ABSTRACT. Aquí va el resumen

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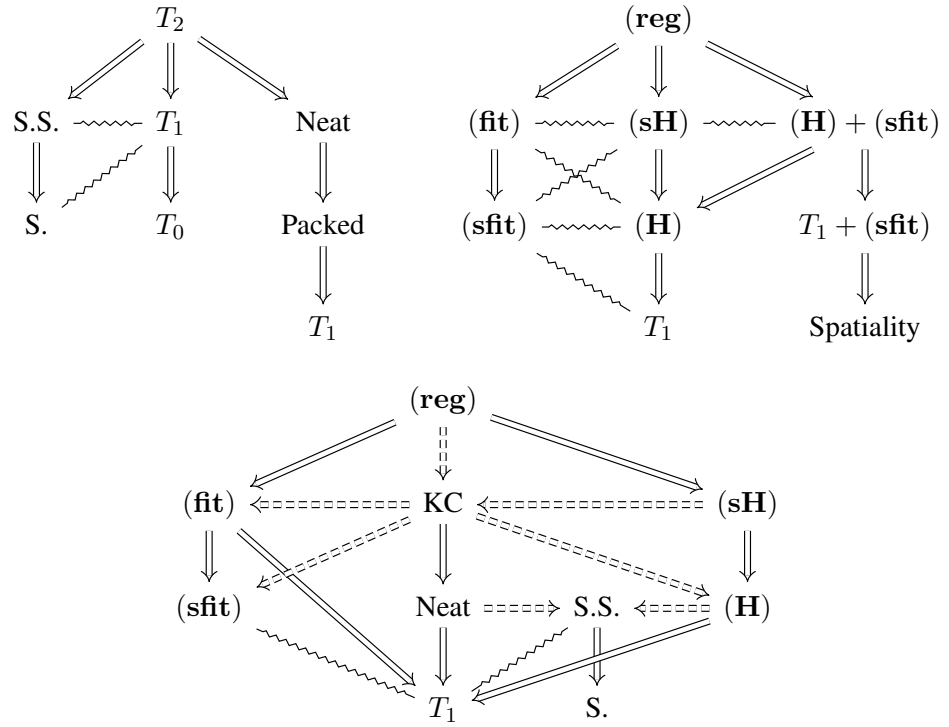
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1. INTRODUCTION

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2. PRELIMINARIES

2.1. **Frames, nuclei, and open filters.** We recall the basic notions and notation concerning frames, nuclei, and open filters; for general background, see the standard references on point-free topology.

A *frame* is a complete lattice A satisfying

$$a \wedge \bigvee X = \bigvee \{a \wedge x \mid x \in X\}.$$

A frame homomorphism preserves finite meets and arbitrary joins. We write $\mathbf{Frm}$ for the category of frames. The Heyting implication on A is characterized by

$$x \wedge a \leq b \quad \Longleftrightarrow \quad x \leq (a \succ b).$$

Every frame homomorphism $f : A \to B$ has a right adjoint $f_* : B \to A$, characterized by $f(a) \leq b$ if and only if $a \leq f_*(b)$.

A *nucleus* on A is a map $j : A \to A$ satisfying

$$x \leq j(x), \qquad j(j(x)) = j(x), \qquad j(x \wedge y) = j(x) \wedge j(y).$$

The fixed-point set $A_j = \{x \in A \mid j(x) = x\}$ is a frame, with joins given by $\bigvee_{A_j} X = j(\bigvee_A X)$. The map $j^* : A \to A_j$, $j^*(x) = j(x)$, is a surjective frame homomorphism, and every quotient arises in this way. We denote by NA the frame of nuclei on A. For $a \in A$, put

$$u_a(x) = a \vee x, \qquad v_a(x) = a \succ x, \qquad w_a(x) = (x \succ a) \succ a.$$

The nuclei u_a and v_a determine the closed and open quotients, respectively.

An element $p \in A$ is $\wedge$-*irreducible* if $p \neq 1$ and $x \wedge y \leq p$ implies $x \leq p$ or $y \leq p$. The set $\mathrm{pt}(A)$ of such elements is endowed with the topology generated by

$$\mathcal{U}(a) = \{p \in \mathrm{pt}(A) \mid a \not\leq p\}, \qquad a \in A.$$

The assignment $\mathcal{U} : A \to \mathcal{O}(\mathrm{pt}(A))$ is a surjective frame homomorphism, called the *spatial reflection*. A frame is *spatial* when $\mathcal{U}$ is an isomorphism. Points correspond bijectively to completely prime filters through $F_p = \{a \in A \mid a \not\leq p\}$.

A filter F on A is *open* if, for every directed $D \subseteq A$,

$$\bigvee D \in F \quad \Longrightarrow \quad D \cap F \neq \varnothing.$$

We write $A^\wedge$ for the collection of open filters. If $f : A \to B$ is a frame homomorphism with right adjoint f_*, then

$$G \in B^\wedge \Longrightarrow f_*G \in A^\wedge, \qquad f_*G = \{a \in A \mid f(a) \in G\}. \tag{1}$$

Proposition 2.1. *Let $f = f^* \colon A \to B$ be a frame morphism. If $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Every nucleus j determines its *admissible filter*

$$\nabla(j) = \{a \in A \mid j(a) = 1\}.$$

Given a filter F, the least nucleus admitting F is

$$v_F = f_F^\infty = \bigvee_{a \in F} v_a, \qquad f_F(x) = \bigvee_{a \in F} (a \succ x), \tag{2}$$

where the join defining v_F is taken in NA. For $F \in A^\wedge$, one has $\nabla(v_F) = F$. Thus, open filters correspond bijectively to fitted nuclei whose admissible filters are open.

The Hofmann–Mislove theorem identifies $A^\wedge$ with the compact saturated subsets of $\mathrm{pt}(A)$. Explicitly, the corresponding sets and filters are

$$Q_F = \{p \in \mathrm{pt}(A) \mid (\forall a \in F)\ a \not\leq p\}, \qquad F_Q = \{a \in A \mid Q \subseteq \mathcal{U}(a)\}.$$

We refer to (F, Q_F) as an ***HM pair***. When $\mathrm{pt}(A) = \varnothing$, the only open filter is A, corresponding to $Q_A = \varnothing$ and to the trivial fitted quotient $v_A = \top_{NA}$.

2.2. **Compact quotients and admissibility intervals.** Recall that the quotient A_j determined by a nucleus j is compact if and only if $\nabla(j)$ is an open filter. In particular, for each $F \in A^\wedge$, the fitted nucleus v_F determines a compact quotient. However, a compact quotient need not be uniquely determined by its admissible filter.

For $F \in A^\wedge$, let

$$M_F = \{m \in A \setminus F \mid m \text{ is maximal in } A \setminus F\}.$$

Define

$$w_F(x) = \bigwedge \{m \in M_F \mid x \leq m\}$$

and the nucleus w_F is the greatest nucleus with admissible filter F, whereas v_F is the least one. Consequently,

$$\{j \in NA \mid \nabla(j) = F\} = [v_F, w_F].$$

We call this the *admissibility interval* associated with F. Evaluating at 0 yields

$$[v_F(0), w_F(0)] \subseteq A.$$

If $\mathrm{pt}(A)$ is T_1, then $M_F = Q_F$, identifying points with their corresponding $\wedge$-irreducible elements.

We shall also use nuclei induced by subspaces. For a topological space S and $E \subseteq S$, define

$$[E] : \mathcal{O}S \longrightarrow \mathcal{O}S, \qquad [E](U) = (E \cup U)^\circ.$$

A nucleus of this form is called *spatially induced*.

Let Q be a compact saturated subset of S, put $G = \{U \in \mathcal{O}S \mid Q \subseteq U\}$, and write $Q' = S \setminus Q$. Then

$$\nabla([Q']) = G, \qquad v_G \leq [Q'] \leq w_G.$$

More generally, let $F \in A^\wedge$, let $Q = Q_F$, and put $G = F_Q = \mathcal{V}(Q)$, viewed as an open filter of $\mathcal{O}(\mathrm{pt}(A))$. We call (F, Q, G) an *HM triple*. Denote by

$$\mathcal{U}_* : \mathcal{O}(\mathrm{pt}(A)) \to A$$

the right adjoint of the spatial reflection.

Proposition 2.2. *For every $k \in [v_G, w_G]$, the composite*

$$\mathcal{U}_* \circ k \circ \mathcal{U}$$

is a nucleus on A with admissible filter F.

Hence Proposition 2.5 gives the map

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

and, in particular,

$$v_F \leq \Phi(k) \leq w_F.$$

Finally, when $A = \mathcal{O}S$, identifying S with $\mathrm{pt}(A)$, the spatial reflection is an isomorphism.

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Lemma 2.3. *If $A = \mathcal{O}S$, then $\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E]$ for every $E \subseteq S$.*

3. Separation Properties Induced by Compact Quotients

The notions studied in this document are largely motivated by the work of Sexton and Simmons on point-sensitive and point-free patch constructions. We are interested in identifying those nuclei associated with open filters that determine compact closed quotients, and in understanding what happens when this requirement is imposed not only on the fitted nuclei v_F, but on other nuclei lying in their corresponding admissibility intervals. We first recall Sexton's original definition of neatness (it is mentioned in [7, Definition 4.3] and [9, Definition 8.2.1]). Besides

providing the starting point for the properties studied in this section, this formulation will make explicit the phenomenon concerning compact closed quotients that motivates our approach.

Definition 3.1. Let A be a frame, let $F \in A^\wedge$ and $\alpha \in \mathrm{Ord}$. We say that:

- F is α-neat, if
$$x \vee d = 1, \quad \text{for every } x \in F,$$
where $d = f_F^\alpha(0)$.
- A is α-neat if every $F \in A^\wedge$ is α-neat.
- A is neat if it is α-neat for some α.

Let us point out a consequence of the preceding definition that will be relevant throughout this work. Suppose that A is α-neat and let $F \in A^\wedge$. If
$$d = f_F^\alpha(0),$$
then $x \vee d = 1$ for every $x \in F$. Hence, u_d admits every element of F, and therefore
$$f_F \leq u_d.$$
Since u_d is a nucleus, it follows that
$$v_F = f_F^\infty \leq u_d.$$
On the other hand, $d = f_F^\alpha(0) \leq f_F^\infty(0) = v_F(0)$, and hence
$$u_d \leq u_{v_F(0)} \leq v_F.$$
Consequently,
$$v_F = f_F^\infty = u_d.$$
Thus, neatness ensures that the fitted nucleus associated with every open filter is a closed nucleus. Since v_F determines a compact quotient, every such quotient is simultaneously compact and closed.

Of particular relevance to our purposes is the spatial characterization of neatness in terms of packed and stacked spaces. More precisely, Sexton proved that a topological space is neat if and only if it is both packed and stacked ([7, Theorem 6.2]). Motivated by the preceding spatial characterization, we now introduce pointfree counterparts of the stacked and strongly stacked conditions.

Let S be a topological space, let $Q \subseteq S$ be compact saturated, and let
$$G = \nabla(Q) \in (\mathcal{O}S)^\wedge$$
be its corresponding open filter. Sexton calls S stacked if
$$v_G(\emptyset) = (Q')^\circ,$$
for every such pair (G, Q), and strongly stacked if
$$v_G = [Q']$$
for every such pair.

We now return to the general setting. Let A be a frame, $S = \mathrm{pt}(A)$, and let (F, Q, G) be an H–M triple. Recall that the map of Proposition (2.2)
$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

provides a natural way of comparing the admissibility interval associated with G with that associated with F.

Proposition 3.2. *Let (F, Q, G) be an H–M triple. Then $\Phi(w_G) = w_F$, that is,*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Proof. By Proposition 2.2, since $w_G \in [v_G, w_G]$, we have

$$\mathcal{U}_* \circ w_G \circ \mathcal{U} \leq w_F.$$

Thus, it remains to prove the reverse inequality.

By adjunction, it is enough to show that

$$\mathcal{U} \circ w_F \leq w_G \circ \mathcal{U}.$$

Let $x \in A$. Since $w_G = [M']$, we have

$$w_G(\mathcal{U}(x)) = [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Hence, we need to prove that

$$\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ.$$

Since $\mathcal{U}(w_F(x))$ is open, it is enough to show that

$$\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x).$$

Suppose, by contradiction, that there exists $y \in \mathcal{U}(w_F(x))$ such that $y \notin M'$ and $y \notin \mathcal{U}(x)$. Then $y \in M$ and $x \leq y$. On the other hand, $y \in \mathcal{U}(w_F(x))$ implies $w_F(x) \not\leq y$. However,

$$w_F(x) = \bigwedge \{m \in M \mid x \leq m\},$$

and since $y \in M$ with $x \leq y$, it follows that $w_F(x) \leq y$ a contradiction.

Therefore, $\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ$. □

The preceding proposition shows that Φ preserves the upper endpoints of the corresponding admissibility intervals:

$$\Phi(w_G) = w_F.$$

Since $[Q']$ is the spatially induced nucleus appearing in Sexton's definitions, its image

$$\Phi([Q']) = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$$

provides the natural candidate for reproducing this behavior on A. This leads to the following definitions.

Definition 3.3. Let A be a frame and (F, Q) be an HM pair. We say that:

- A is FQ-*stacked* if

$$(U_* \circ [Q'] \circ U)(0) = v_F(0).$$

- A is *stacked* if it is FQ-stacked for every HM pair (F, Q).

Definition 3.4. Let A be a frame and (F, Q) be an HM pair. We say that:

- A is FQ-*strongly stacked* if
$$U_* \circ [Q'] \circ U = v_F.$$
- A is *strongly stacked* if it is FQ-strongly stacked for every HM pair (F, Q).

Clearly, every strongly stacked frame is stacked. Let us look that these properties are conservative. We first establish the relation between the transfinite iterations defining the fitted nuclei on a space and on its frame of open sets.

Lemma 3.5. *Let $\alpha \in \mathrm{Ord}$ and $X \in CS$. Then*
$$\partial_F^\alpha(X) = \left(f_F^\alpha(X')\right)'.$$
In particular, for every $U \in \mathcal{O}S$,
$$\left(\partial_F^\infty(U')\right)' = v_F(U).$$

Proof. We proceed by transfinite induction on α.

For $\alpha = 0$, both ∂_F^0 and f_F^0 are the identity. Hence,
$$\partial_F^0(X) = X = (X')' = \left(f_F^0(X')\right)'.$$

Suppose that the result holds for some ordinal α. Then
$$\begin{aligned}
\partial_F^{\alpha+1}(X) &= \partial_F\left(\partial_F^\alpha(X)\right) \\
&= \left(f_F\left((\partial_F^\alpha(X))'\right)\right)' \\
&= \left(f_F\left(f_F^\alpha(X')\right)\right)' \\
&= \left(f_F^{\alpha+1}(X')\right)'.
\end{aligned}$$

Finally, let λ be a limit ordinal and suppose that the result holds for every $\alpha < \lambda$. By definition and De Morgan's law,
$$\begin{aligned}
\left(\partial_F^\lambda(X)\right)' &= \left(\bigcap_{\alpha<\lambda} \partial_F^\alpha(X)\right)' \\
&= \bigcup_{\alpha<\lambda} (\partial_F^\alpha(X))' \\
&= \bigcup_{\alpha<\lambda} f_F^\alpha(X') \\
&= f_F^\lambda(X'),
\end{aligned}$$
where the third equality follows from the induction hypothesis. Therefore,
$$\partial_F^\lambda(X) = \left(f_F^\lambda(X')\right)'.$$
This completes the proof. □

We are now in a position to prove the conservativity of both notions.

Proposition 3.6. *The notions of stackedness and strong stackedness are conservative.*

Proof. Let $S \in \mathrm{Top}$ and set $A = \mathcal{O}S$.

First, suppose that S is strongly stacked. Then, for every HM pair (F, Q),

$$v_F = [Q'].$$

By Lemma 2.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F,$$

and hence $\mathcal{O}S$ is strongly stacked.

Now, suppose that S is stacked. Then, for every HM pair (F, Q),

$$Q = \partial_F^\infty(S).$$

Hence,

$$Q' = \left(\partial_F^\infty(S)\right)'.$$

By Lemma 3.5,

$$v_F(\varnothing) = \left(\partial_F^\infty(S)\right)' = Q'.$$

In particular, $Q' \in \mathcal{O}S$. By Lemma 2.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\begin{aligned}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(\varnothing) &= [Q'](\varnothing) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\varnothing).\end{aligned}$$

Thus, $\mathcal{O}S$ is stacked.

Since all the implications used above are equivalences, the converse statements follow by reversing the argument. □

Recall that neatness requires the fitted nucleus associated with each open filter to determine a closed compact quotient. It is then natural to ask what happens if this requirement is extended to every nucleus admitting an open filter. This leads to a point-free counterpart of the classical KC condition considered by Wilansky [12], where compact subsets are required to be closed. Accordingly, we introduce the following notion.

Definition 3.7. Let A be a frame. We say that A is *KC* if every compact quotient of A is closed.

Equivalently, A is KC if, for every $j \in NA$ such that

$$\nabla(j) \in A^\wedge,$$

there exists $a \in A$ such that

$$j = u_a.$$

In the spatial setting, Wilansky's KC condition coincides with Rosemary's notion of packedness. Moreover, in this case one has

$$\text{Neat} \iff (\text{Packed} = \text{KC}) \text{ and Stacked.}$$

This naturally raises the question of whether analogous relationships hold among the point-free notions introduced in this section. We shall return to this question in Section 4. Before doing so, we first examine how these notions interact with the classical separation axioms in Frm.

3.1. **Comparison with Classical Separation Axioms.** Using the conservativity of the stacked and strongly stacked properties, one can already observe that neither of these properties is comparable with the T_1 separation axiom. Indeed, there exist strongly stacked (and hence stacked) non-spatial frames which are not T_1, as well as T_1 non-spatial frames which are not stacked (and hence not strongly stacked).

Example 3.8.

(1) Consider the frame $A = [0,1]$ with its usual order. Its open filters are of the form
$$F_r = (r,1], \qquad 0 \leq r < 1,$$
and the corresponding compact saturated subsets of $\mathrm{pt}\, A = [0,1)$ are
$$Q_r = [0,r].$$
A direct computation shows that
$$v_{F_r}(x) = \begin{cases} x, & x \leq r, \\ 1, & x > r, \end{cases} = (\mathcal{U}_* \circ [Q_r'] \circ \mathcal{U})(x).$$
Hence, A is strongly stacked. However, A is not T_1.

(2) The Hausdorff frame of Paseka and Šmarda [4] considered in Example (3.16) is T_1, but it is not stacked. Consequently, it is not strongly stacked either.

Recall that if a topological space S is T_1 and strongly stacked, then
$$w_G = v_G.$$
where $G \in \mathcal{O}S^{\wedge}$. Moreover, every T_1 stacked space is sober. This naturally raises the question of whether analogous results hold in the point-free setting. In this section, we investigate this question.

Proposition 3.9. *Let A be a frame. If*
$$v_F(0) = w_F(0)$$
for every $F \in A^{\wedge}$, then every point of A is maximal in $\mathrm{pt}\, A$.

Proof. Let $p, q \in \mathrm{pt}\, A$ be such that $p \leq q$, and consider the open filter
$$F_q = \{x \in A \mid x \not\leq q\}.$$
Since $A \setminus F_q = {\downarrow} q$, it follows directly from the definition of w_{F_q} that
$$w_{F_q} = w_q. \tag{3}$$

In particular,

$$w_{F_q}(0) = q.$$

By hypothesis,

$$q = w_{F_q}(0) = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every $x \in F_q$, we have $x \not\leq q$. Since $p \leq q$, it follows that $x \not\leq p$. Moreover,

$$x \wedge \neg x = 0 \leq p.$$

Since p is a point, it is meet-irreducible, and therefore

$$\neg x \leq p.$$

Thus,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x \leq p.$$

Since $p \leq q$, we conclude that $p = q$. Hence, every point of A is maximal in $\operatorname{pt} A$. □

Proposition 3.10. *Let A be a frame. The following statements hold:*

(1) *If A is T_1 and stacked, then*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$.

(2) *If*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then A is stacked.

(3) *Suppose that*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$ and that

$$\mathcal{V}(a) = \{q \in \operatorname{pt} A \mid a \leq q\} \neq \varnothing$$

for every $a \neq 1 \in A$. Then A is T_1.

Proof.

(1) Let (F, Q) be an HM pair. Since A is T_1, the space $S = \operatorname{pt} A$ is T_1. Hence $Q = M$. Since A is stacked,

$$\begin{aligned} v_F(0) &= (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \\ &= (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) \\ &= w_F(0). \end{aligned}$$

(2) Let (F, Q) be an HM pair. Recall that

$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F. \tag{4}$$

Evaluating at 0, we obtain

$$v_F(0) \leq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \leq w_F(0).$$

By hypothesis, $v_F(0) = w_F(0)$, and hence

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0).$$

Therefore, A is stacked.

(3) Let $p \in \operatorname{pt} A$ and $a \in A$ be such that

$$p \leq a.$$

By hypothesis, there exists $q \in \operatorname{pt} A$ such that

$$a \leq q.$$

Hence,

$$p \leq a \leq q.$$

By Proposition 3.9, every point of A is maximal. Therefore, $p = q$, and consequently

$$p \leq a \leq p.$$

Thus, $a = p$, and hence A is T_1.

□

Thus, in the T_1 case, the stacked condition forces the lower and upper bounds of every admissibility interval to agree at 0. The strongly stacked case yields the corresponding equality of the nuclei themselves.

Corollary 3.11. *Let A be a frame. The following statements hold:*

(1) *If A is T_1 and strongly stacked, then*

$$v_F = w_F$$

for every $F \in A^\wedge$.

(2) *If*

$$v_F = w_F$$

for every $F \in A^\wedge$, then A is strongly stacked.

(3) *Suppose that*

$$v_F = w_F$$

for every $F \in A^\wedge$ and that

$$\mathcal{V}(a) = \{q \in \operatorname{pt} A \mid a \leq q\} \neq \varnothing$$

for every $a \in A$. Then A is T_1.

Proof.

(1) Let (F, Q) be an HM pair. Since A is strongly stacked,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Moreover, since A is T_1, we have $Q = M$. Hence, by Proposition 3.2,

$$v_F = \mathcal{U}_* \circ [M'] \circ \mathcal{U} = w_F.$$

(2) By (4),
$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$
Thus, if $v_F = w_F$, then
$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U},$$
and therefore A is strongly stacked.

(3) Since $v_F = w_F$ implies
$$v_F(0) = w_F(0),$$
the result follows from Proposition 3.10 (3).

□

While the strongly stacked condition yields the collapse of every admissibility interval, the stacked condition already guarantees this phenomenon for those associated with completely prime filters.

Corollary 3.12. *Let A be a T_1 stacked frame. Then every admissibility interval associated with a completely prime filter collapses to a singleton. More precisely, for every $p \in \operatorname{pt} A$,*
$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

Proof. Let $p \in \operatorname{pt} A$. Since A is T_1 and stacked, Proposition 3.10 gives
$$v_{F_p}(0) = w_{F_p}(0) = p.$$
Moreover, as p is maximal,
$$w_{F_p} = w_p = u_p.$$
Recall that, for every nucleus j, one has
$$u_{j(0)} \leq j.$$
Indeed, for every $x \in A$,
$$j(0) \vee x \leq j(x).$$
Applying this to $j = v_{F_p}$ yields
$$u_p = u_{v_{F_p}(0)} \leq v_{F_p}.$$
On the other hand, by definition of the admissibility interval,
$$v_{F_p} \leq w_{F_p} = u_p.$$
Therefore,
$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

□

Notice that the additional hypothesis in Proposition 3.10 (3) is automatically satisfied whenever A is spatial. Indeed, if $a < 1$ and A is spatial, then there exists $p \in \operatorname{pt} A$ such that
$$a \leq p.$$
Hence,
$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing.$$

Thus, in the spatial setting, Proposition 3.15 reflects the topological fact that every T_1 stacked space is sober.

The point-free situation is, however, genuinely broader, since the T_1 and stacked conditions do not force spatiality.

Example 3.13. Let B be an atomless complete Boolean algebra and consider

$$A = B \times 2,$$

where $2 = \{0 < 1\}$. Then A is a complete Boolean algebra. We show that A is T_1 and strongly stacked, but not spatial.

First, let

$$p = (1, 0).$$

We claim that

$$\operatorname{pt} A = \{p\}.$$

Indeed, suppose that $(a, b), (c, d) \in A$ satisfy

$$(a, b) \wedge (c, d) \leq p.$$

Then

$$(a \wedge c, b \wedge d) \leq (1, 0),$$

so $b \wedge d = 0$. Since $b, d \in 2$, either $b = 0$ or $d = 0$, and hence

$$(a, b) \leq p \qquad \text{or} \qquad (c, d) \leq p.$$

Thus, $p \in \operatorname{pt} A$.

Now, let $q = (x, y) \in \operatorname{pt} A$. If $y = 0$, then

$$(1, 0) \wedge (x, 1) \leq (x, 0) = q.$$

Since $(x, 1) \not\leq q$, the complete meet-irreducibility of q implies

$$(1, 0) \leq q,$$

and therefore $x = 1$. Hence $q = p$.

If $y = 1$, then $x < 1$. For $a, b \in B$ such that $a \wedge b \leq x$, we have

$$(a, 1) \wedge (b, 1) \leq (x, 1) = q.$$

Since q is a point, either $a \leq x$ or $b \leq x$, showing that $x \in \operatorname{pt} B$. This is impossible, since an atomless complete Boolean algebra has no points. Therefore,

$$\operatorname{pt} A = \{p\}.$$

Moreover, if

$$p \leq (a, b),$$

then $a = 1$, and hence (a, b) is either p or 1. Thus p is maximal and A is T_1.

On the other hand, A is not spatial. Indeed, if $a, b \in B$ with $a \neq b$, then

$$(a, 0) \neq (b, 0),$$

while

$$\mathcal{U}(a, 0) = \varnothing = \mathcal{U}(b, 0).$$

Hence the spatial reflection

$$\mathcal{U}\colon A \longrightarrow \mathcal{O}(\operatorname{pt} A)$$

is not injective.

It remains to prove that A is strongly stacked. Since B is an atomless complete Boolean algebra, 0 is its only compact element. Therefore,

$$K(A) = K(B) \times K(2) = \{(0,0),(0,1)\}.$$

Since every open filter of a complete Boolean algebra is principal and $\uparrow a$ is open if and only if a is compact, it follows that

$$A^{\wedge} = \{\uparrow (0,0), \uparrow (0,1)\}.$$

Clearly,

$$\uparrow (0,0) = A,$$

whereas

$$\uparrow (0,1) = B \times \{1\} = F_p.$$

Hence,

$$A^{\wedge} = \{A, F_p\}.$$

For F_p, since A is T_1, then

$$v_{F_p} = w_{F_p}.$$

For $F = A$, we have

$$v_F = \operatorname{tp} = w_F.$$

Consequently,

$$v_F = w_F$$

for every $F \in A^{\wedge}$. By Corollary 3.11, A is strongly stacked.

Theorem 3.14. *There exist non-spatial T_1 strongly stacked frames. In particular, there exist non-spatial T_1 stacked frames.*

Recall that $A^{\wedge}$ is endowed with the topology generated by the basic open sets

$$D(x) = \{F \in A^{\wedge} \mid x \in F\}, \qquad x \in A.$$

Moreover, let VA denote the Vietoris modification of A introduced in [3]. The points of VA are of the form

$$(F,a), \qquad F \in A^{\wedge}, \quad a \in [v_F(0), w_F(0)].$$

Consider the continuous surjection

$$\pi\colon \operatorname{pt}(VA) \longrightarrow A^{\wedge}, \qquad (F,a) \longmapsto F.$$

It induces the frame morphism

$$\mathcal{O}(\pi) = \pi^{-1}\colon \mathcal{O}(A^{\wedge}) \longrightarrow \mathcal{O}(\operatorname{pt}(VA)).$$

Recall also that the subbasic open sets of $\operatorname{pt}(VA)$ are determined by

$$(F,a) \in \Box x \iff x \in F, \qquad (F,a) \in \Diamond x \iff x \not\leq a. \tag{5}$$

Particularly,

$$a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F\rangle(x) \iff x \not\leq a. \tag{6}$$

Theorem 3.15. *Let A be a frame, and consider the continuous surjection*

$$\pi\colon \operatorname{pt}(VA) \longrightarrow A^{\wedge}, \qquad (F, a) \longmapsto F.$$

Then the following statements hold:

(1) *If A is T_1 and stacked, then*

$$\mathcal{O}(\pi) = \pi^{-1}\colon \mathcal{O}(A^{\wedge}) \longrightarrow \mathcal{O}(\operatorname{pt}(VA))$$

is an isomorphism.

(2) *If $\mathcal{O}(\pi)$ is an isomorphism, then A is stacked.*

(3) *If $\mathcal{O}(\pi)$ is an isomorphism and*

$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing$$

for every $a < 1$, then A is T_1.

Proof.

(1) Suppose that A is T_1 and stacked. By Proposition 3.10,

$$v_F(0) = w_F(0)$$

for every $F \in A^{\wedge}$. Hence, each fiber of

$$\pi\colon \operatorname{pt}(VA) \longrightarrow A^{\wedge}$$

contains a unique element. Therefore, π is bijective.

It remains to prove that

$$\pi^{-1}\colon A^{\wedge} \longrightarrow \operatorname{pt}(VA)$$

is continuous. Recall that the subbasic open sets of $\operatorname{pt}(VA)$ are given by

$$(F, a) \in [-](x) \iff x \in F$$

and

$$(F, a) \in \langle - \rangle(x) \iff x \not\leq a.$$

For the first family,

$$(F, a) \in [-](x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi([-](x)) = D(x).$$

Now consider $\langle - \rangle(x)$. Since the fibers of π collapse, for every $(F, a) \in \operatorname{pt}(VA)$ we have

$$a = v_F(0).$$

Thus,

$$\begin{aligned}(F, a) \notin \langle - \rangle(x) &\iff x \leq a \\ &\iff x \leq v_F(0) \\ &\iff v_F(x) = v_F(0) \\ &\iff x \notin F.\end{aligned}$$

Hence,

$$(F, a) \in \langle - \rangle(x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi(\langle - \rangle(x)) = D(x).$$

Thus, the image under π of every subbasic open set of $\mathrm{pt}(VA)$ is open in $A^\wedge$. Hence π^{-1} is continuous, and consequently π is a homeomorphism. Therefore,

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA))$$

is an isomorphism.

(2) Suppose that

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA))$$

is an isomorphism. Since π^{-1} is bijective, π is injective. For every $F \in A^\wedge$,

$$(F, v_F(0)), (F, w_F(0)) \in \mathrm{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

Hence,

$$v_F(0) = w_F(0).$$

By Proposition 3.10, A is stacked.

(3) Suppose that $\mathcal{O}(\pi)$ is an isomorphism and that

$$\mathcal{V}(a) = \{q \in \mathrm{pt}\, A \mid a \leq q\} \neq \varnothing$$

for every $a < 1$. By part (2),

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$. Therefore, by Proposition 3.9, every point of A is maximal.

Let $p \in \mathrm{pt}\, A$ and $a \in A$ be such that

$$p \leq a.$$

By hypothesis, there exists $q \in \mathrm{pt}\, A$ such that

$$a \leq q.$$

Thus,

$$p \leq a \leq q.$$

Since p and q are maximal, we have $p = q$, and therefore

$$p \leq a \leq p.$$

Hence $a = p$, and A is T_1.

□

We now compare stackedness with the stronger separation property (**H**). As shown in [4], property (**H**) implies T_1. Nevertheless, property (**H**) does not imply stackedness, as the following example shows.

Example 3.16. There exists a frame satisfying property (**H**) which is not stacked.

Let $I = [0, 1]$, endowed with its usual topology, and consider the frame

$$A = \mathcal{O}(I).$$

Since I is a compact Hausdorff space, A is a compact Hausdorff frame. Moreover, since I is T_1, every point of A is maximal and

$$D(A) = \operatorname{pt} A = \{I \setminus \{x\} \mid x \in I\},$$

where $D(A)$ denotes the set of maximal elements of A.

Following the construction of Paseka and Šmarda [4, Proposition 2.3], let

$$\widetilde{A} = \{(u, v) \in A \times A_{\neg\neg} \mid u \leq v\},$$

where

$$A_{\neg\neg} = \{a \in A \mid w_0(a) = a\}$$

is the Boolean algebra of regular elements of A. By [4, Propositions 2.3 and 2.4], $\widetilde{A}$ is a compact frame. In particular, $\widetilde{A}$ satisfies property (**H**).

The maximal elements of $\widetilde{A}$ are precisely the elements of the form

$$D(\widetilde{A}) = \{(d, 1) \mid d \in D(A)\}.$$

Since $\widetilde{A}$ satisfies (**H**), it is T_1, and hence its points are precisely its maximal elements. Therefore,

$$\operatorname{pt} \widetilde{A} = \{(I \setminus \{x\}, I) : x \in I\}.$$

Since $\widetilde{A}$ is compact,

$$F = {\uparrow} 1 = \{1\}$$

is an open filter. Let (F, Q) be the corresponding HM pair. By the Hofmann–Mislove correspondence,

$$Q = \operatorname{pt} \widetilde{A} - F.$$

Since $1 \notin \operatorname{pt} \widetilde{A}$, it follows that

$$Q = \operatorname{pt} \widetilde{A}.$$

Now, since $F = \{1\}$,

$$f_F = v_1 = \mathrm{id}_{\widetilde{A}},$$

and therefore

$$v_F(0) = 0_{\widetilde{A}}.$$

On the other hand,

$$\begin{aligned}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) &= \bigwedge Q \\ &= \bigwedge \operatorname{pt} \widetilde{A} \\ &= (\varnothing, I) \neq (\varnothing, \varnothing) = v_F(0).\end{aligned}$$

Thus,

$$v_F(0) \neq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0),$$

and consequently $\widetilde{A}$ is not stacked.

At this point, it is useful to consider another point-free version of the Hausdorff separation axiom. A frame A is said to be strongly Hausdorff, or simply ($\mathbf{sH}$), if the codiagonal

$$\nabla_A : A \oplus A \longrightarrow A$$

is a closed quotient. This stronger condition has an important consequence for the compact quotients considered here.

Proposition 3.17. *Every strongly Hausdorff frame is KC.*

Proof. This follows directly from of [6, Theorem 9.2.1], which states that every compact quotient of a strongly Hausdorff frame is closed. □

Proposition 3.18. *Every KC frame is T_1.*

Proof. Let $p \in \mathrm{pt}\, A$. The filter

$$F_p = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$$

is completely prime and, in particular, $F_p \in A^\wedge$. Hence A_{w_p} is a compact quotient of A. Since A is KC, this quotient is closed. Therefore,

$$w_p = u_a$$

for some $a \in A$. Evaluating at 0, we obtain

$$a = u_a(0) = w_p(0) = p,$$

and hence

$$w_p = u_p.$$

Now let $a \in A$ be such that

$$p < a \leq 1.$$

Since $a \not\leq p$, we have

$$w_p(a) = 1.$$

On the other hand,

$$u_p(a) = p \vee a = a.$$

Since $w_p = u_p$, it follows that

$$a = 1.$$

Thus p is maximal. Since every point of A is maximal, A is T_1. □

We briefly recall two separation properties that will be useful in what follows. A frame A is said to be fit if, whenever $a \not\leq b$, there exist $x, y \in A$ such that

$$a \vee x = 1, \qquad y \not\leq b, \qquad x \wedge y \leq b.$$

On the other hand, A is said to be subfit if, whenever $a \not\leq b$, there exists $c \in A$ such that

$$a \vee c = 1 \qquad \text{and} \qquad b \vee c \neq 1.$$

Every fit frame is subfit. We will denote these properties by ($\mathbf{fit}$) and ($\mathbf{sfit}$), respectively.

Corollary 3.19. *Every subfit KC frame is spatial.*

Proof. Let A be a subfit KC frame. Since every KC frame is T_1, the result follows from [6, Proposition 2.5.3]. □

Theorem 3.20. *There exists a KC frame which is strongly stacked and not subfit. Consequently,*

$$\mathrm{KC} \not\Rightarrow (\mathrm{sfit}).$$

Proof. Consider the frame

$$A = B \times 2,$$

introduced in Example 3.13. Recall that A is T_1, strongly stacked, and non-spatial.

The only open filters of A are A and $F_p = {\uparrow}(0,1)$, where $p = (1,0)$ is the unique point of A.

For $F = A$, we have

$$v_A = w_A = \mathrm{tp} = u_1.$$

On the other hand, for $F = F_p$, by Corollary 3.12 yield

$$v_{F_p} = w_{F_p} = u_p.$$

Thus, for every $F \in A^\wedge$, every nucleus in the admissibility interval $[v_F, w_F]$ is closed. Hence A is a KC frame.

Finally, A cannot be subfit. Indeed, every T_1 subfit frame is spatial, whereas A is T_1 and non-spatial. Consequently, A is a KC frame which is strongly stacked and not subfit. In particular,

$$\mathrm{KC} \not\Rightarrow (\mathrm{sfit}).$$

□

3.2. **Inheritance and preservation properties.** The notions of stacked and strongly stacked frames introduced above are formulated in terms of the admissibility intervals associated with Hofmann–Mislove pairs. A closely related phenomenon had already appeared in the work of Simmons.

More precisely, Simmons shows in [10, Lemma 8.9 and Corollary 8.10] that, for a Hofmann–Mislove pair (F, Q), with $G = \nabla(Q)$, the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u} \\ \mathcal{O}S & \xrightarrow[v_G]{} & \mathcal{O}S \end{array} \tag{7}$$

commutes laxly; that is,

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Here

$$v_F = f_F^\infty \qquad \text{and} \qquad v_G = f_G^\infty$$

are the fitted nuclei associated with $F \in A^\wedge$ and $G \in (\mathcal{O}S)^\wedge$, respectively.

In the terminology developed in the preceding sections, Simmons' inequality may be viewed as an early manifestation of the behavior underlying our notions of stackedness: it provides a canonical comparison between the lower bounds of the corresponding admissibility intervals in A and $\mathcal{O}S$.

We now show that this phenomenon is not restricted to the spatial reflection. In fact, an analogous comparison holds for an arbitrary frame morphism. Let

$$h\colon A \longrightarrow B$$

be a frame morphism, let $G \in B^\wedge$, and set

$$F = h_*G \in A^\wedge.$$

We shall compare the fitted nuclei determined by these filters through the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h} \\ B & \xrightarrow[v_G]{} & B. \end{array}$$

As in Simmons' situation, the relevant comparison is first established at the level of the prenuclei whose transfinite iterations generate the corresponding fitted nuclei. The following lemma provides the basic inequality needed for this purpose.

Lemma 3.21. *Let $h\colon A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_*G$. Then*

$$h \circ f_F \leq f_G \circ h.$$

Proof. Recall that, by (1),

$$F = h_*G = \{a \in A \mid h(a) \in G\}.$$

Hence

$$f_F = \bigvee\{v_a \mid a \in F\} \qquad \text{and} \qquad f_G = \bigvee\{v_y \mid y \in G\}.$$

Let $x \in A$. For every $a \in F$, we have $h(a) \in G$, and therefore

$$h(v_a(x)) = h(a \to x) \leq h(a) \to h(x) = v_{h(a)}(h(x)) \leq f_G(h(x)).$$

Thus, since h preserves arbitrary joins,

$$\begin{aligned} h(f_F(x)) &= h\left(\bigvee_{a\in F} v_a(x)\right) \\ &= \bigvee_{a\in F} h(v_a(x)) \\ &\leq f_G(h(x)). \end{aligned}$$

Hence

$$h \circ f_F \leq f_G \circ h.$$

□

The preceding inequality is preserved throughout the transfinite iteration of the corresponding prenuclei.

Lemma 3.22. *Let $h\colon A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_*G$. Then, for every ordinal α,*

$$h \circ f_F^\alpha \leq f_G^\alpha \circ h.$$

In particular,

$$h \circ v_F \leq v_G \circ h.$$

Proof. We proceed by transfinite induction on α. If $\alpha = 0$, the result is immediate.

For the successor step, suppose that the inequality holds for α. Then, by Lemma 3.21 and the induction hypothesis,

$$\begin{aligned} h \circ f_F^{\alpha+1} &= h \circ f_F \circ f_F^{\alpha} \\ &\leq f_G \circ h \circ f_F^{\alpha} \\ &\leq f_G \circ f_G^{\alpha} \circ h \\ &= f_G^{\alpha+1} \circ h. \end{aligned}$$

Finally, let λ be a limit ordinal. For every $a \in A$, since h preserves arbitrary joins, we have

$$\begin{aligned} h\big(f_F^{\lambda}(a)\big) &= h\left(\bigvee_{\alpha<\lambda} f_F^{\alpha}(a)\right) \\ &= \bigvee_{\alpha<\lambda} h\big(f_F^{\alpha}(a)\big) \\ &\leq \bigvee_{\alpha<\lambda} f_G^{\alpha}(h(a)) \\ &= f_G^{\lambda}(h(a)), \end{aligned}$$

where the inequality follows from the induction hypothesis. Hence

$$h \circ f_F^{\lambda} \leq f_G^{\lambda} \circ h.$$

Since

$$v_F = f_F^{\infty} \qquad \text{and} \qquad v_G = f_G^{\infty},$$

taking an ordinal at which both transfinite iterations have stabilized yields

$$h \circ v_F \leq v_G \circ h.$$

This completes the proof. □

Proposition 3.23. *If A is a neat frame, then A_j is a neat frame.*

Proof. Suppose that A is α-neat. Let $G \in A_j^{\wedge}$ and set

$$F = j_* G \in A^{\wedge}.$$

Consider the corresponding prenuclei

$$f_F = \bigvee_{x \in F} v_x \qquad \text{and} \qquad f_G = \bigvee_{y \in G} v_y.$$

Since A is α-neat, for every $x \in F$,

$$x \vee \widehat{d} = 1, \qquad \text{where } \widehat{d} = f_F^{\alpha}(0).$$

Set

$$d = f_G^{\alpha}(0).$$

By Proposition 3.22, applied to the frame morphism $j = j^* \colon A \to A_j$,

$$j(\widehat{d}) = j\big(f_F^{\alpha}(0)\big) \leq f_G^{\alpha}(j(0)) = f_G^{\alpha}(0) = d.$$

Since j is inflationary,

$$\widehat{d} \leq j(\widehat{d}) \leq d.$$

Now let $y \in G$. Since $y \in A_j$, we have $j(y) = y$. Thus, by (1),

$$y \in F = j_* G.$$

Consequently,

$$y \vee \widehat{d} = 1.$$

Since $\widehat{d} \leq d$, it follows that

$$1 = y \vee \widehat{d} \leq y \vee d \leq 1.$$

Hence

$$y \vee d = 1.$$

Therefore, A_j is α-neat, and hence neat. □

The following proposition establishes that neatness is preserved under binary coproducts.

Proposition 3.24. *Let A_1 and A_2 be α-neat frames. Then $A_1 \oplus A_2$ is α-neat.*

Proof. Let

$$\iota_i \colon A_i \longrightarrow A_1 \oplus A_2, \qquad i = 1, 2,$$

be the canonical frame morphisms, and let

$$F \in (A_1 \oplus A_2)^{\wedge}.$$

For $i = 1, 2$, set

$$F_i = (\iota_i)_* F \in A_i^{\wedge}.$$

Since A_i is α-neat, if

$$d_i = f_{F_i}^{\alpha}(0),$$

then

$$x_i \vee d_i = 1$$

for every $x_i \in F_i$.

By the definition of $F_i = (\iota_i)_* F$,

$$x_i \in F_i \quad \Longleftrightarrow \quad \iota_i(x_i) \in F.$$

Under the usual representation of the binary coproduct, this gives

$$x_1 \oplus 1, \ 1 \oplus x_2 \in F.$$

Since F is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now set

$$d = f_F^{\alpha}(0 \oplus 0).$$

By Proposition 3.22, applied to the canonical morphisms ι_i, we have

$$\iota_i \circ f_{F_i}^{\alpha} \leq f_F^{\alpha} \circ \iota_i, \qquad i = 1, 2.$$

Evaluating at $0 \in A_i$, we obtain

$$\iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \qquad \text{and} \qquad \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &= \iota_1(d_1) \wedge \iota_2(d_2) \\ &\leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) \\ &= f_F^\alpha\big((0 \oplus 1) \wedge (1 \oplus 0)\big) \\ &= f_F^\alpha(0 \oplus 0) \\ &= d. \end{aligned}$$

Finally, for $x_i \in F_i$,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Hence

$$(x_1 \oplus x_2) \vee d = 1 \oplus 1.$$

Thus $A_1 \oplus A_2$ is α-neat. □

Corollary 3.25. *Let A_1 and A_2 be α-neat and β-neat, respectively. Then $A_1 \oplus A_2$ is γ-neat, where*

$$\gamma = \max\{\alpha, \beta\}.$$

Proof. Let $\gamma = \max\{\alpha, \beta\}$. Then A_1 and A_2 are both γ-neat. Hence, by Proposition 3.24, $A_1 \oplus A_2$ is γ-neat. □

We extend the previous result to arbitrary coproducts.

Theorem 3.26. *Neatness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of neat frames, and for each $i \in I$, let α_i be a neatness degree of A_i. Consider

$$A = \bigoplus_{i \in I} A_i$$

and let

$$\iota_i \colon A_i \longrightarrow A$$

denote the canonical coproduct morphisms.

Let $F \in A^\wedge$. For each $i \in I$, set

$$F_i = (\iota_i)_* F \in A_i^\wedge.$$

Moreover,

$$x_i \in F_i \quad \Longleftrightarrow \quad \iota_i(x_i) \in F.$$

Set

$$\alpha = \sup\{\alpha_i \mid i \in I\}.$$

Then every A_i is α-neat. Hence, for every $x_i \in F_i$,

$$x_i \vee d_i = 1,$$

where

$$d_i = f^{\alpha}_{F_i}(0).$$

Let

$$f_F = \bigvee_{x \in F} v_x$$

be the prenucleus associated with F. By Proposition 3.22, applied to each canonical coproduct morphism $\iota_i \colon A_i \to A$, we have

$$\iota_i \circ f^{\alpha}_{F_i} \leq f^{\alpha}_{F} \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain

$$\iota_i(d_i) = \iota_i\big(f^{\alpha}_{F_i}(0)\big) \leq f^{\alpha}_{F}(\iota_i(0)) = f^{\alpha}_{F}(\langle 0 \rangle).$$

Set

$$d = f^{\alpha}_{F}(\langle 0 \rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every $i \in I$. Taking arbitrary joins yields

$$\langle d_i \rangle \vee \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since $x_i \vee d_i = 1$ for every $i \in I$, we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As $\langle d_i \rangle \leq d$,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore,

$$A = \bigoplus_{i \in I} A_i$$

is α-neat, and hence neat. □

The preceding results have a categorical consequence. Indeed, the subcategory of neat frames is closed under quotients, by Proposition 3.23, and under arbitrary coproducts, by Proposition 3.26. Hence, the standard criterion for coreflectivity of full subcategories of Frm applies.

Corollary 3.27. *The subcategory of neat frames is a coreflective subcategory of* Frm.

We now turn to the inheritance of the stronger properties introduced above. We first show that strong stackedness is inherited by quotients, and then establish the analogous result for the KC property.

Recall that, for a frame morphism $f\colon A \to B$, $j \in NA$, and $k \in NB$, we have

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NB$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* denotes the right adjoint of f.

Proposition 3.28. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is strongly stacked.*

Proof. Let $F \in A_j^{\wedge}$, and set

$$S = \operatorname{pt} A, \qquad S_j = \operatorname{pt} A_j.$$

Consider the open filter

$$G = j_*(F) = \{x \in A \mid j(x) \in F\},$$

and let

$$Q = S_j \setminus F, \qquad R = S \setminus G.$$

Denote by

$$\mathcal{U}\colon A \longrightarrow \mathcal{O}S \qquad \text{and} \qquad \mathcal{U}_j\colon A_j \longrightarrow \mathcal{O}S_j$$

the corresponding spatial reflections.

Set

$$l = N(\mathcal{U}_j)_*([Q']) = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j$$

and

$$m = N(j^*)(v_G).$$

By Proposition 2.2,

$$v_F \leq l.$$

Thus, it is enough to prove

$$l \leq m \leq v_F.$$

We first prove that $m \leq v_F$. Since $G = j_*(F)$, the fitted nuclei associated with G and F satisfy

$$j^* \circ v_G \leq v_F \circ j^*. \tag{8}$$

Moreover,

$$m = N(j^*)(v_G) = \bigvee_{x \in A} \left(u_{j^*(v_G(x))} \wedge v_{j^*(x)}\right).$$

By (8), for every $x \in A$,

$$j^*(v_G(x)) \leq v_F(j^*(x)),$$

and hence

$$u_{j^*(v_G(x))} \leq u_{v_F(j^*(x))}.$$

Therefore,

$$u_{j^*(v_G(x))} \wedge v_{j^*(x)} \leq u_{v_F(j^*(x))} \wedge v_{j^*(x)} \leq v_F.$$

Taking joins over $x \in A$, we obtain

$$m \leq v_F. \tag{9}$$

We now prove that $l \leq m$. Since A is strongly stacked,

$$v_G = N(\mathcal{U})_*([R']) = \mathcal{U}_* \circ [R'] \circ \mathcal{U}.$$

Hence,

$$m = N(j^*)\big(N(\mathcal{U})_*([R'])\big).$$

To compare the u-components in the decompositions of l and m, consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)_*([Q']).$$

We first observe that, for every $x \in A_j$,

$$j^*(n(x)) = l(x).$$

Indeed,

$$\begin{aligned} j^*(n(x)) &= j^* \left(j_* \circ (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j \circ j^*\right)(x) \\ &= j^* j_* \left(\bigvee \left\{a \in A_j \;\middle|\; \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j^*(x))\right\}\right) \\ &= \bigvee \left\{a \in A_j \;\middle|\; \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(x)\right\} \\ &= l(x). \end{aligned}$$

Now, under the usual identification of S_j with the corresponding subspace of S, we have

$$Q = R \cap S_j.$$

Using the decomposition of a nucleus in terms of its u- and v-components, we have

$$l = \bigvee_{x \in A_j} \left(u_{l(x)} \wedge v_x\right) = \bigvee_{x \in A_j} \left(u_{j^*(n(x))} \wedge v_x\right).$$

On the other hand, since

$$m = N(j^*)\big(N(\mathcal{U})_*([R'])\big),$$

we have

$$m = \bigvee_{a \in A} \left(u_{j^*(N(\mathcal{U})_*([R'])(a))} \wedge v_{j^*(a)}\right).$$

For $x \in A_j$, the explicit description of the right adjoint gives

$$\begin{aligned} j^*\big(N(\mathcal{U})_*([R'])(x)\big) &= j^* \left(\bigvee \left\{a \in A \;\middle|\; \mathcal{U}(a) \subseteq R' \cup \mathcal{U}(x)\right\}\right) \\ &= \bigvee \left\{j^*(a) \in A_j \;\middle|\; \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\right\}. \end{aligned}$$

Moreover, since $x \in A_j$,

$$\mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)$$

is equivalent, after restriction to S_j, to

$$\mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(x).$$

Consequently,

$$j^*(n(x)) \leq j^*\big(N(\mathcal{U})_*([R'])(x)\big).$$

Therefore,

$$u_{j^*(n(x))} \leq u_{j^*(N(\mathcal{U})_*([R'])(x))}$$

for every $x \in A_j$, and hence

$$u_{l(x)} \wedge v_x = u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})_*([R'])(x))} \wedge v_x \leq m.$$

Taking joins over $x \in A_j$, we obtain

$$l \leq m.$$

Combining this inequality with (9) and $v_F \leq l$, we obtain

$$v_F \leq l \leq m \leq v_F.$$

Hence,

$$v_F = l = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j.$$

Therefore, A_j is strongly stacked. □

In particular, since every strongly stacked frame is stacked, we also obtain that A_j is stacked.

We next show that the KC property is also inherited by quotients.

Proposition 3.29. *Let A be a KC frame and $j \in NA$. Then A_j is KC.*

Proof. Let $k \in N(A_j)$ be such that $(A_j)_k$ is compact. Then

$$\nabla(k) \in A_j^{\wedge}.$$

By Proposition 2.1,

$$j_*(\nabla(k)) \in A^{\wedge}.$$

Consider the nucleus

$$l = j_* \circ k \circ j^* \in NA.$$

Since

$$\nabla(l) = j_*(\nabla(k)),$$

the quotient A_l is compact. Since A is KC, there exists $a \in A$ such that

$$l = u_a.$$

We claim that

$$k = u_{j^*(a)}.$$

Indeed, from $l = u_a$ we have

$$a \vee x = l(x) = j_*\big(k(j^*(x))\big)$$

for every $x \in A$. Applying j^* to both sides yields

$$j^*(a) \vee j^*(x) = k(j^*(x)),$$

since $j^* \circ j_* = \mathrm{id}_{A_j}$.

Now let $x \in A_j$. Since $j^*\colon A \to A_j$ is surjective, we have $j^*(x) = x$, and therefore

$$\begin{aligned} u_{j^*(a)}(x) &= j^*(a) \vee x \\ &= j^*(a) \vee j^*(x) \\ &= k(j^*(x)) \\ &= k(x). \end{aligned}$$

Hence

$$k = u_{j^*(a)}.$$

Thus $(A_j)_k$ is closed. Therefore, A_j is KC. □

We conclude this section by examining more closely the relation between the compact quotients of a frame and those arising from its quotients. Besides the inheritance results established above, the comparison obtained in Corollary 3.22 induces a natural frame morphism between the corresponding compact quotients.

By Proposition 3.22, $j \circ v_F \leq v_G \circ j$. We now compare the compact quotients induced by F and G. Consider

$$A_F = \{x \in A \mid v_F(x) = x\} \qquad \text{and} \qquad A_{j,G} = \{y \in A_j \mid v_G(y) = y\}.$$

Define

$$H\colon A \longrightarrow A_{j,G}, \qquad H = v_G \circ j.$$

Since v_F is inflationary,

$$v_G \circ j \leq v_G \circ j \circ v_F.$$

On the other hand, by Proposition 3.22 and the idempotency of v_G,

$$v_G \circ j \circ v_F \leq v_G \circ v_G \circ j = v_G \circ j.$$

Hence

$$H \circ v_F = H.$$

Therefore, H factors through the quotient A_F, inducing a map

$$h\colon A_F \longrightarrow A_{j,G}$$

such that

$$h \circ v_F = v_G \circ j.$$

Explicitly,

$$h(x) = v_G(j(x)), \qquad x \in A_F.$$

Lemma 3.30. *The map*

$$h\colon A_F \longrightarrow A_{j,G}$$

preserves arbitrary joins.

Proof. Let $X \subseteq A_F$. Since joins in A_F are given by

$$\bigvee_{A_F} X = v_F\left(\bigvee_{A} X\right),$$

we have

$$\begin{aligned} h\left(\bigvee_{A_F} X\right) &= v_G\left(j\left(v_F\left(\bigvee_{A} X\right)\right)\right) \\ &\leq v_G\left(v_G\left(j\left(\bigvee_{A} X\right)\right)\right) \\ &= v_G\left(j\left(\bigvee_{A} X\right)\right), \end{aligned}$$

where the inequality follows from Proposition 3.22 and the equality from the idempotency of v_G.

Since

$$H = v_G \circ j \colon A \longrightarrow A_{j,G}$$

is a frame morphism,

$$v_G\left(j\left(\bigvee_{A} X\right)\right) = \bigvee_{A_{j,G}} \{H(x) \mid x \in X\} = \bigvee_{A_{j,G}} h[X].$$

Therefore,

$$h\left(\bigvee_{A_F} X\right) \leq \bigvee_{A_{j,G}} h[X].$$

Conversely, for every $x \in X$,

$$x \leq \bigvee_{A_F} X,$$

and hence, by monotonicity of h,

$$h(x) \leq h\left(\bigvee_{A_F} X\right).$$

Since this holds for every $x \in X$,

$$\bigvee_{A_{j,G}} h[X] \leq h\left(\bigvee_{A_F} X\right).$$

Thus,

$$h\left(\bigvee_{A_F} X\right) = \bigvee_{A_{j,G}} h[X].$$

Therefore, h preserves arbitrary joins. □

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 3.31. *The diagram*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle h} \\ A_j & \xrightarrow[v_G]{} & A_{j,G} \end{array} \tag{10}$$

is commutative.

Proof. By Lemma 3.30, h is a frame morphism. Moreover, by the definition of h,

$$h \circ v_F = v_G \circ j.$$

Hence the diagram commutes. □

We now specialize the construction obtained above to the spatial reflection. Let A be a frame and set $S = \operatorname{pt} A$. For $F \in A^\wedge$, let $Q \in \mathcal{Q}S$ be the compact saturated subset associated with F by the Hofmann–Mislove correspondence, and put $G = \nabla(Q) \in (\mathcal{O}S)^\wedge$.

Taking $j = \mathcal{U}_A \colon A \longrightarrow \mathcal{O}S$ in diagram (10), we obtain the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle h} \\ \mathcal{O}S & \xrightarrow[v_G]{} & (\mathcal{O}S)_G. \end{array}$$

By [9, Lemma 5.7.1], the point space of the compact quotient A_F is precisely Q; that is,

$$\operatorname{pt}(A_F) = Q.$$

Hence, the spatial reflection of A_F is

$$\mathcal{U}_{A_F} \colon A_F \longrightarrow \mathcal{O} \operatorname{pt}(A_F) = \mathcal{O}Q.$$

On the other hand, since $G = \nabla(Q)$, the points of the compact quotient $(\mathcal{O}S)_G$ are precisely the points of Q. Thus,

$$\operatorname{pt}((\mathcal{O}S)_G) = Q,$$

and its spatial reflection is

$$\mathcal{U}_{(\mathcal{O}S)_G} \colon (\mathcal{O}S)_G \longrightarrow \mathcal{O}Q.$$

Consequently, the preceding constructions fit into the following commutative diagram:

$$\begin{array}{ccccc} A & \xrightarrow{v_F} & A_F & & \\ {\scriptstyle \mathcal{U}_A}\downarrow & & {\scriptstyle h}\downarrow & \searrow^{\mathcal{U}_{A_F}} & \\ & & & & \mathcal{O}Q \\ & & & \nearrow_{\mathcal{U}_{\mathcal{O}S_\nabla}} & \\ \mathcal{O}S & \xrightarrow[v_\nabla]{} & \mathcal{O}S_\nabla & & \end{array} \tag{11}$$

Since this diagram is determined by $F \in A^{\wedge}$ and its associated compact saturated subset $Q \in \mathcal{Q}S$, we shall refer to it as the Q-square.

Although property $(\mathbf{H})$ does not in general guarantee that compact quotients are closed, the particular compact quotient appearing in (11) exhibits a closely related behavior.

Theorem 3.32. *Under the hypotheses of the Q-square, if A satisfies property $(\mathbf{H})$, then*

$$\mathcal{O}Q \cong\uparrow Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies $(\mathbf{H})$. Since $(\mathbf{H})$ is conservative, $S = \operatorname{pt} A$ is a T_2 space. Hence, for every $Q \in \mathcal{Q}S$, the set Q is closed, and therefore $Q' \in \mathcal{O}S$.

Moreover, since every T_2 space is strongly stacked, we have

$$v_{\nabla} = [Q'].$$

As $Q' \in \mathcal{O}S$, it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_{\nabla}} = (\mathcal{O}S)_{u_{Q'}} \cong\uparrow Q'.$$

Now, $u_{Q'}$ is a complemented nucleus in $\mathcal{O}S$, by [5, Proposition VI.3.3], implies that $\mathcal{O}S_{v_{\nabla}}$ is spatial. Therefore,

$$(\mathcal{O}S)_{v_{\nabla}} \cong \mathcal{O}\operatorname{pt}((\mathcal{O}S)_{v_{\nabla}}).$$

Since

$$\operatorname{pt}((\mathcal{O}S)_{v_{\nabla}}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_{\nabla}} \cong\uparrow Q'.$$

□

4. Interaction among Compact-Quotient Separation Properties

We begin by recalling a relation established by Sexton in the spatial setting. By [7, Theorem 6.2 (a)], for every topological space S,

$$\mathcal{O}S \text{ neat} \implies S \text{ strongly stacked} \implies S \text{ stacked}.$$

The first of these implications provides a natural starting point for comparing neatness and strong stackedness in the point-free setting. Indeed, although neatness of a frame does not directly imply strong stackedness of the frame itself, it does imply strong stackedness of its point space.

Proposition 4.1. *If A is a neat frame, then $\operatorname{pt} A$ is strongly stacked.*

Proof. Let $S = \operatorname{pt} A$. Since neatness is hereditary, $\mathcal{O}S$ is neat. By [7, Theorem 6.2 (a)], S is strongly stacked. □

DONE: Ejemplo de neat no KC

DONE: Comprobar si neat implica stacked

DONE: KC implica S.S.?

Existe alguna propiedad P tal que (H)+P implica staked

T1+S.+P implica (H)?

DONE: Relacionar S. y S.S. con fit y sfit

The preceding proposition concerns the point space of A, rather than the frame A itself. It is therefore natural to ask under what additional conditions neatness of A implies that A is strongly stacked. The comparison induced by the spatial reflection provides such a condition. For every HM triple (F, Q, G), we have

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Suppose now that A is strongly stacked. Then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since

$$v_G \leq [Q'],$$

we obtain

$$\mathcal{U}_* \circ v_G \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F.$$

Together with the previous inequality, this yields

$$v_F = \mathcal{U}_* \circ v_G \circ \mathcal{U},$$

or equivalently,

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}.$$

Thus, strong stackedness forces the comparison induced by the spatial reflection to commute. Under neatness, the converse also holds.

Proposition 4.2. *Let A be a neat frame. If*

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$$

for every HM triple (F, Q, ∇), then A is strongly stacked.

Proof. Let $S = \operatorname{pt} A$ and let (F, Q, ∇) be an HM triple.

Since A is neat, $\mathcal{O}S$ is neat. Hence S is stacked and packed. In particular, every compact saturated subset $Q \subseteq S$ is closed, so

$$Q' \in \mathcal{O}S.$$

Since S is stacked,

$$Q = \partial_\nabla^\infty(S),$$

and therefore, by Lemma 3.5,

$$v_\nabla(\varnothing) = Q'.$$

Because $\mathcal{O}S$ is neat, it follows that

$$v_\nabla = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U} = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint $\mathcal{U}_*$, we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore, A is strongly stacked. □

We next turn to the KC property and compare it with the preceding notions. We first observe that KC always implies neatness.

Proposition 4.3. *Let A be a KC-frame. Then A is neat.*

Proof. Let $F \in A^\wedge$ and put

$$d = v_F(0).$$

Since A is KC and

$$v_F \in [v_F, w_F],$$

there exists $a \in A$ such that

$$v_F = u_a.$$

Evaluating at 0, we obtain

$$d = v_F(0) = u_a(0) = a.$$

Hence

$$v_F = u_d.$$

Since this holds for every $F \in A^\wedge$, the frame A is neat. □

Proposition 4.4. *Let A be a KC frame. Then A is stacked if and only if it is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since every KC-frame is T_1, we have

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Moreover, by the previous proposition,

$$v_F = u_d.$$

Since A is KC, w_F is closed as well, and therefore

$$w_F = u_a.$$

Consequently,

$$\begin{aligned} A \text{ is strongly stacked} &\iff v_F = w_F \\ &\iff u_d = u_a \\ &\iff d = a \\ &\iff v_F(0) = w_F(0) \\ &\iff A \text{ is stacked.} \end{aligned}$$

Since this holds for every HM-pair (F, Q), the result follows. □

The preceding proposition does not assert that every KC-frame is stacked. In fact, the following example shows that KC does not imply even stackedness.

Example 4.5.

Proposition 4.6. *Let A be a spatial frame. Then A is KC if and only if it is neat.*

Proof. Suppose first that A is KC. By the previous results, every KC frame is neat.

Conversely, let A be neat. Since A is spatial, we may write

$$A \cong \mathcal{O}(S)$$

for some topological space S. By the spatial characterization of neatnees, S is packed and stacked. Moreover, neatness implies that S is strongly stacked. Hence S is packed and strongly stacked, and therefore $\mathcal{O}(S)$ is KC. Thus A is KC. □

The preceding results show that KC imposes strong restrictions on the relation between neatness and the two forms of stackedness. We now consider converses under additional separation hypotheses.

Proposition 4.7. *Let A be a KC frame. If A is subfit, then A is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since A is KC, both v_F and w_F are closed nuclei. Hence

$$v_F = u_d \qquad \text{and} \qquad w_F = u_a.$$

By subfitness and [11, Theorem 3.5], we have

$$d = a.$$

Therefore,

$$v_F = u_d = u_a = w_F.$$

Since A is T_1, we also have

$$w_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Consequently,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since this holds for every HM pair (F, Q), A is strongly stacked. □

Proposition 4.8. *Let A be a frame. If A is neat and fit, then A is KC.*

Proof. Let $F \in A^\wedge$ and $j \in NA$ be such that

$$\nabla(j) = F.$$

Since A is fit, by [11, Theorem 3.4], every nucleus on A is fitted. Hence j is the least nucleus in its block. On the other hand, v_F is also the least nucleus admitting F. Therefore,

$$j = v_F.$$

Since A is neat, there exists $d \in A$ such that

$$v_F = u_d.$$

Consequently,

$$j = u_d.$$

Thus every nucleus whose associated filter is open is closed, and hence A is KC. □

Proposition 4.9. *Every neat strongly stacked frame is KC.*

Proof. Let $F \in A^\wedge$. Since A is neat, it is T_1. Hence, as A is strongly stacked, we have

$$v_F = w_F.$$

Moreover, of neatness,

$$v_F = u_d, \qquad d = v_F(0).$$

Therefore,

$$[v_F, w_F] = \{u_d\}.$$

Since every nucleus inducing a compact quotient of A belongs to an admissibility interval $[v_F, w_F]$ for some $F \in A^\wedge$, every such nucleus is closed. Hence every compact quotient of A is closed, and therefore A is KC. □

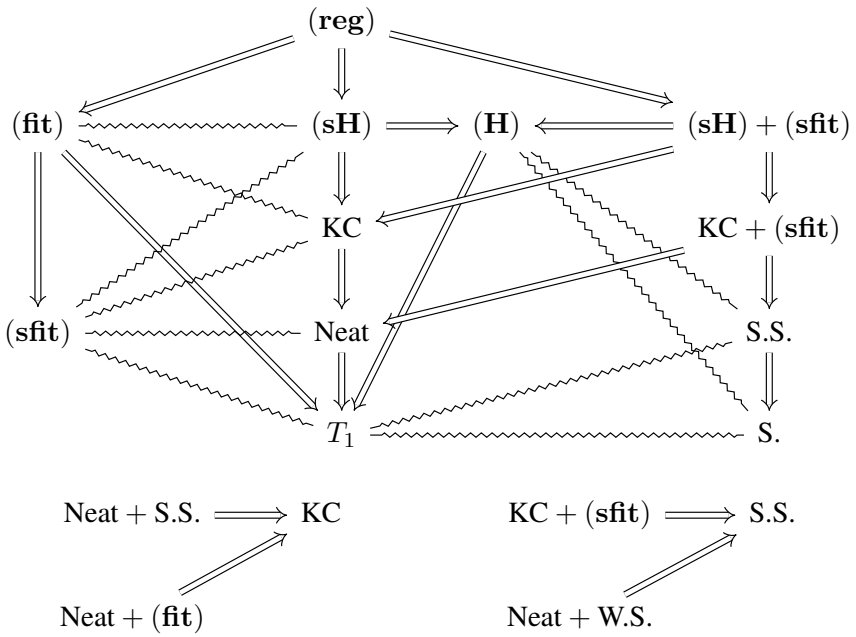
Proposition 4.6 shows that neatness and the KC property coincide for spatial frames. However, this equivalence does not extend to arbitrary frames. The following theorem provides a non-spatial neat frame admitting a compact quotient which is not closed, thereby establishing that the implication from KC to neatness is strict in the point-free setting.

Theorem 4.10. *There exists a non-spatial neat frame which is not KC. Consequently, the implication*

$$\mathrm{KC} \Longrightarrow \mathrm{Neat}$$

is strict in the category of frames.

Proof. □



5. Relationship with the patch assembly

Following the idea presented in [11] (or, alternatively, in [10]), let us consider the linearly ordered frame

$$A = [0,1].$$

In this case the following observations hold.

(1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

(2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases} \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

(3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a,1] \quad \text{o} \quad F = \nabla(v_a) = [a,1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a,1]$.

(4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases} \tag{12}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

(5) For $S = \operatorname{pt} A$ it holds that $S = [0,1)$ and the open sets in $\mathcal{O}S$ are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0,x).$$

(6) The frame A is not T_1, since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm.

(7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0,m]$.

(8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.

(9) Since $[0,x) \in \mathcal{O}S$, then $[x,1] \in \mathcal{C}S$ and $Q = [0,m]$. Therefore, the compact subsets are not closed; that is, S is not packed.

(10) The frame A is not efficient.

The above resume some characteristics about the frame $[0,1]$. Now, we will perform the analysis of the patch constructions of A and its point space S.

Remember that for $A \in \mathrm{Frm}$ and $S \in \mathrm{Top}$ we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S. By (12) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0,1], m \in [0,1)\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m.

Case 1: If $x < m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x. We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then

$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a), where $m \in \operatorname{pt} A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \operatorname{pt} A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc}
A & \longrightarrow & PA & \hookrightarrow & NA \\
{\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle P(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\
\mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\
& \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\
& & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S
\end{array}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S. In [8] is named *the patch assembly diagram* and more details about it can also be found in [**?**].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong P\mathcal{O}S$. Similarly, $NA \cong N\mathcal{O}S$. To describe explicitly $P\mathcal{O}S$ and $N\mathcal{O}S$, we use the following result, which can be consulted in [1].

Proposition 5.1. *Let S be a partially ordered set.*

(1) *If S has no infinite antichains, then $N\mathcal{O}S$ is spatial.*
(2) *If S is totally ordered, then $N\mathcal{O}S$ is spatial.*

Thus, by Proposition 5.1 we have that $N\mathcal{O}S$ is spatial and therefore $N\mathcal{O}S \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong\downarrow & & \downarrow\cong & & \downarrow\cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame PA, we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0,a) \cap [m,1) \mid a \in [0,1], m \in [0,1)\} = \{(m,a)\}.$$

Since the topology generated by the intervals (m,a) coincides with the usual topology on $[0,1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies $(\mathbf{H})$, even when A is not T_1 nor neat.

Lemma 5.2. *Let A be a frame, if the morphism $\mathrm{p}U \colon \mathrm{p}A \to \mathrm{p}\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q, thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we neee to observe the action of $\mathrm{p}U$ on these two nuclei, in order to do that remember that we have a conmutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & \mathrm{p}A & \overset{\iota}{\hookrightarrow} & NA \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle \mathrm{p}(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\ \mathcal{O}S & \longrightarrow & \mathrm{p}\mathcal{O}S & \overset{\imath}{\hookrightarrow} & N\mathcal{O}S \\ & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = \imath \mathrm{p}\mathcal{U}$, that is, $\mathrm{p}\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to $\mathrm{p}A$, then

$$\mathrm{p}\mathcal{U}(\mathrm{v}_F) = N(\mathcal{U})(\mathrm{v}_F) = \bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee\{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $\mathrm{v}_{\mathcal{U}F}$.

Completar esta prueba

Considerar incluir información sobre los parches

□

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